DATA ANALYTICS

WEATHER DATA ANALYSIS USING POWER BI DASHBOARD

An Internship Report Submitted

In partial fulfilment of the requirement for the award of the degree of

BACHELOR OF COMPUTER SCIENCE

SUBMITTED

By

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PG & RESEARCH DEPARTMENT OF COMPUTER SCIENCE

SCHOOL OF MATHEMATICAL COMPUTATION SCIENCES

HOLY CROSS COLLEGE (AUTONOMOUS)

Affiliated To Bharathidasan University

Nationally Accredited (4th Cycle) With ‘A++’ Grade (CGPA 3.75/4) By NAAC

College With Potential For Excellence

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BONAFIDE CERTIFICATE

This is to certify that the Internship Program on report submitted in partial fulfillment for the award of the Bachelor of Computer Science is the record of work done by Hasna Nachiyar S , U24CS043 of the Department of Computer Science, Holy Cross College (Autonomous), Tiruchirappalli during the period of her study in the academic year 2025-2026.

Submitted for the Viva-Voce Examination held on ______________________________

Signature of the Staff In-Charge **Signature of the Head of the Department**

Signature of the External Examiner

ACKNOWLEDGEMENT

"**YOU HAVE ALL YOU NEED IN MY STRENGTH".** I thank LORD JESUS for his showers of blessings and for his divine who enabled me to complete this Internship Program successfully. This work has been undertaken and completed with direct and indirect help of many people and I would like to acknowledge the same.

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HASNA NACHIYAR S

HITAKEY TECH SOLUTION PVT LTD

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THIS INTERNSHIP CERTIFICATE

HITAKEY

is presented to

HASNA NACHIYAR S

from BSC. CS HOLY CROSS COLLEGE dept/ College

for completing internship on DATA ANALYTICS

26-12-2025 to 12-01-2026

During this period, the intern actively participated in assigned tasks, demonstrated professionalism, and contributed to the organization with dedication and enthusiasm.

For HITAKEY TECH SOLUTION PVT LTD.

Director

DIRECTOR

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OBJECTIVES OF THE INTERNSHIP

The main objectives of the Data Analytics internship are:

- To understand the basic concepts of data analytics
- To learn how to clean and organize datasets
- To develop skills in Microsoft Excel for data analysis
- To create visual reports and dashboards using Power BI
- To perform a real-time hands-on data analysis project
- To gain practical knowledge of business data analysis

TOOLS USED

Microsoft Excel

Microsoft Excel is one of the most widely used tools for data analysis. It allows users to store, organize, calculate, and visualize data easily.

Key features learned:

- Data entry and formatting
- Sorting and filtering data
- Conditional formatting
- Excel formulas (SUM, AVERAGE, COUNT, IF, VLOOKUP)
- Pivot Tables
- Charts and graphs

Excel helped in performing basic data cleaning and initial analysis before moving the data to Power BI.

POWER BI

Power BI is a business intelligence tool developed by Microsoft used for data visualization and interactive dashboards.

Key features learned:

- Data import and transformation
- Power Query Editor
- Data modelling
- Creating charts and dashboards
- DAX formulas
- Interactive report creation

Power BI helps convert raw data into meaningful visual insights.

CHAPTER 1
INTRODUCTION

Data Analytics is the process of examining raw data to discover useful patterns, trends, and insights that help organizations make better decisions. In today's digital world, businesses generate large amounts of data from various sources such as sales transactions, customer feedback, and online platforms. Analyzing this data helps organizations understand customer behavior, improve efficiency, and increase profitability.

This internship program focused on learning the fundamentals of data analytics using industry-standard tools such as Microsoft Excel and Power BI. The training included both theoretical concepts and practical exercises that helped develop data analysis and visualization skills. Excel was used for data cleaning, sorting, filtering, and performing calculations, while Power BI was used to create interactive dashboards and reports.

During the internship, a hands-on project was completed to apply the concepts learned during the training. The project involved analyzing a dataset, performing data transformation, and presenting the results using visual dashboards. This internship helped in gaining practical knowledge of data analytics tools and understanding how data-driven decision making works in real-world organizations

CHAPTER 2
BASIC OF MICROSOFT EXCEL

Microsoft Excel is a powerful spreadsheet application developed by Microsoft that is widely used for organizing, storing, and analyzing data in a structured format. It allows users to enter information in the form of rows and columns, creating cells where different types of data such as text, numbers, dates, and formulas can be stored. Excel provides a variety of built-in functions and tools that help users perform calculations quickly and accurately. For example, functions like SUM, AVERAGE, COUNT, MAX, and MIN make it easy to analyze numerical data without performing manual calculations. In addition to calculations, Excel also offers features such as sorting and filtering, which allow users to arrange and view specific data easily. Another important feature of Excel is data visualization, where users can create charts, graphs, and tables to represent information clearly and effectively. These visual tools help in understanding patterns, trends, and comparisons in data. Excel also supports formatting options such as changing fonts, adding colors, applying borders, and highlighting important information to make the data more readable and organized. Because of these capabilities, Microsoft Excel is widely used in various fields including business, finance, education, research, and data analysis. It helps individuals and organizations manage large amounts of data efficiently, make better decisions based on data insights, and present information in a professional manner. Overall, Microsoft Excel plays an important role as a basic and essential tool for data management, calculation, and analysis in modern workplaces and academic environments.

CHAPTER 3
EXCEL FUNCTION AND DATA CLEANING

EXCEL FUNCTION

Excel functions are built-in formulas in Microsoft Excel that help users perform calculations and analyze data quickly and accurately. These functions simplify complex calculations by automatically processing numerical and textual data entered in cells. Some commonly used functions include SUM, which adds a range of numbers; AVERAGE, which calculates the mean value; COUNT, which counts the number of cells containing numbers; and MAX and MIN, which find the highest and lowest values in a dataset. Excel functions are widely used in data analysis, financial calculations, and reporting because they save time, reduce errors and improve efficiency when working with large amounts of data.

DATA CLEANING

Before building the dashboard, the dataset was cleaned.

Tasks performed:

• Removed duplicate records
• Filled missing values
• Corrected data formatting
• Verified numeric and date fields

CHAPTER 4
PIVOT TABLE AND CHART CREATION

Pivot Tables

Pivot tables were created to summarize and organize the dataset effectively. They help in analysing large amounts of data by grouping and calculating values in a structured format.

The pivot tables included summaries such as:

• Data grouped by different categories

• Category-wise comparison of values

• Time-based performance analysis

• Identification of top records

Pivot tables help users quickly summarize large datasets and identify important patterns or trends.

Chart Creation

Charts were created from the pivot tables to visualize the summarized results. These charts help in presenting the data in a clearer and more understandable format.

The charts used include:

• Column Chart – for comparing values across categories

• Pie Chart – for showing percentage distribution

• Line Chart – for displaying trends over time

• Bar Chart – for comparing different items or groups

These visualizations make the data easier to interpret and support better analysis and decision-making.

CHAPTER 5
INTRODUCTION TO POWER BI

Power BI is a powerful business intelligence and data visualization tool developed by Microsoft that helps users collect, analyze, and present data in a clear and interactive manner. It allows users to connect to various data sources such as Excel files, databases, cloud services, and online platforms, making it easier to gather information from multiple places in one system. Power BI provides tools to clean, transform, and organize raw data into meaningful information that can be easily understood. One of the main features of Power BI is its ability to create interactive dashboards and reports using visual elements such as charts, graphs, maps, tables, and cards. These visualizations help users quickly identify trends, patterns, and relationships within the data. Power BI also allows users to update and refresh data automatically, ensuring that the information displayed is always current and accurate. Another advantage of Power BI is that it supports data sharing and collaboration, enabling teams and organizations to access reports and insights easily. It is widely used in businesses, finance, healthcare, education, and many other industries to support data-driven decision making. By transforming complex data into simple visual insights, Power BI helps organizations understand their performance, monitor progress, and make better strategic decisions. Overall, Power BI is an essential tool for data analysis and visualization in modern workplaces.

CHAPTER 6
DATA COLLECTION AND DATASET DESCRIPTION

The data used in this project was collected from a publicly available weather dataset that contains various weather-related parameters. The dataset includes information such as temperature, humidity, pressure, evaporation, sunshine duration, cloud cover, and rainfall details. This data helps in analyzing weather patterns and understanding how different climatic factors affect rainfall and temperature conditions.

The dataset consists of multiple records representing daily weather observations. Each record contains several attributes such as maximum temperature, minimum temperature, rainfall, humidity levels, cloud cover, and wind direction. These variables are useful for studying relationships between different weather conditions and for creating visualizations and dashboards to analyze trends and patterns in the weather data.

Before performing analysis, the dataset was carefully examined and organized to ensure accuracy and consistency. Missing values and inconsistencies were identified and handled during the data preparation stage. This step helped improve the quality of the data and ensured that it was suitable for further analysis and visualization in the dashboard.

CHAPTER 7
DATA IMPORT AND DATA MODELLING

The dataset prepared in Excel was imported into Microsoft Power BI to perform data analysis and create visual dashboards. Power BI provides options to connect and import data from different sources such as Excel files, CSV files, databases, and online services. After importing the dataset, the data was reviewed to ensure that all columns and records were correctly loaded. This step helps in confirming that the dataset is ready for further processing and visualization.

Once the data was imported, basic data transformation was performed using the Power Query Editor in Power BI. During this stage, unnecessary columns were removed, missing values were handled, and data types were corrected to maintain consistency. These transformations help improve the quality of the dataset and make it suitable for accurate analysis.

Data modeling is an important step that organizes the dataset in a structured way for analysis. In this project, fields such as temperature, rainfall, humidity, evaporation, and pressure were properly arranged to support different types of visualizations. If multiple tables are used, relationships can be created between them to ensure data integrity. A well-designed data model improves performance and makes it easier to create meaningful reports.

After organizing the data, calculated fields and measures can be created to derive additional insights from the dataset. These measures help in calculating averages, totals, and comparisons that are used in charts and dashboard visuals. This step enhances the analytical capability of the report and allows users to explore the data more effectively through interactive visualizations.

CHAPTER 8
DASHBOARD CREATION IN POWER BI

The dataset was first organized and processed using **Microsoft Excel**. Various Excel formulas were used to perform calculations and summarize the data. These calculations helped in preparing the dataset for visualization and further analysis.
Some commonly used formulas included:

• **SUM()** – used to calculate totals
• **AVERAGE()** – used to find the mean value
• **COUNT()** – used to count the number of records
• **Basic arithmetic formulas** for performing calculations

After completing the calculations, different charts and key metrics were arranged on a **single worksheet to create a dashboard**. The dashboard was designed to display important information clearly and help users understand the data in a simple and organized manner.
The dashboard included the following elements:

• Key Performance Indicator (KPI) cards
• Comparison charts
• Trend analysis charts
• Category-based charts
• Summary tables

To make the dashboard more interactive, **slicers and filters** were added so that users could view data based on different conditions.
• Category filters
• Time-based filters
• Other relevant filter options

These filters allowed users to explore the data dynamically and made the dashboard more **interactive and user-friendly**.

DATA VISUALIZATION IN POWER BI

After cleaning and preparing the dataset, it was imported into **Microsoft Power BI** to create advanced visualizations. Power BI was used to design interactive dashboards that help in understanding patterns and trends within the data.
Some of the visualizations created in Power BI include:

- Bar and column charts
- Line charts for trend analysis
- Pie or donut charts for category comparison
- Tables and matrix visuals for detailed information

These visual reports made it easier to **analyze data, identify patterns, and interpret information effectively**, helping users gain meaningful insights from the dataset.

CHAPTER 9
PROJECT

This project presents an interactive **Weather Analytics Dashboard using Power BI** to analyze key weather parameters such as temperature, rainfall, sunshine, evaporation, and atmospheric pressure. The dashboard visualizes weather data to identify patterns, relationships, and trends that influence rainfall and climate conditions.

The report includes various visualizations such as KPI cards, scatter plots, bar charts, donut charts, and gauges to summarize important insights like average, minimum, and maximum temperature, the number of rainy days, and the distribution of rainfall predictions. It also explores relationships between variables such as sunshine vs evaporation, cloud cover vs atmospheric pressure, and minimum vs maximum temperature based on rainfall conditions.

By transforming raw weather data into meaningful visuals, the dashboard helps users monitor weather trends, analyze climate behavior, and support data-driven decision-making for environmental analysis and forecasting.

CHAPTER 10
OUTPUT SCREENSHOTS

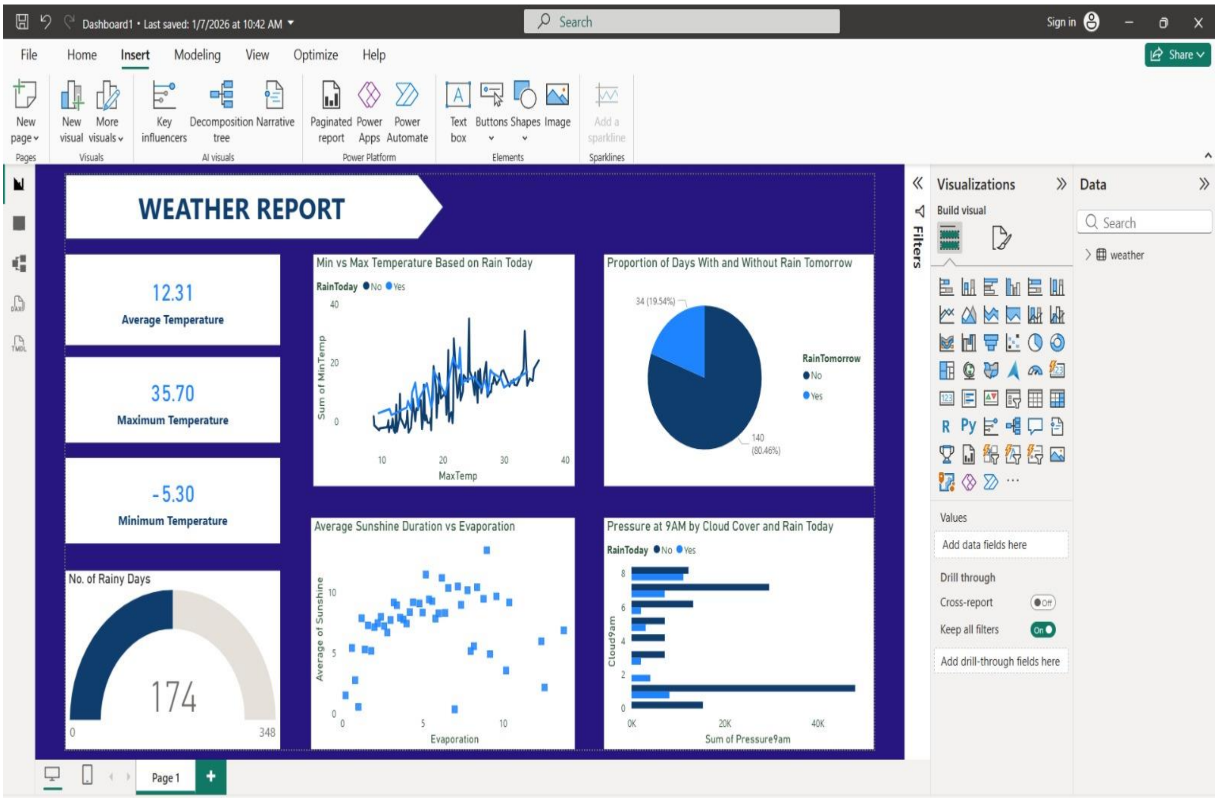

Figure 1

The image shows a weather analysis dashboard created in Microsoft Power BI

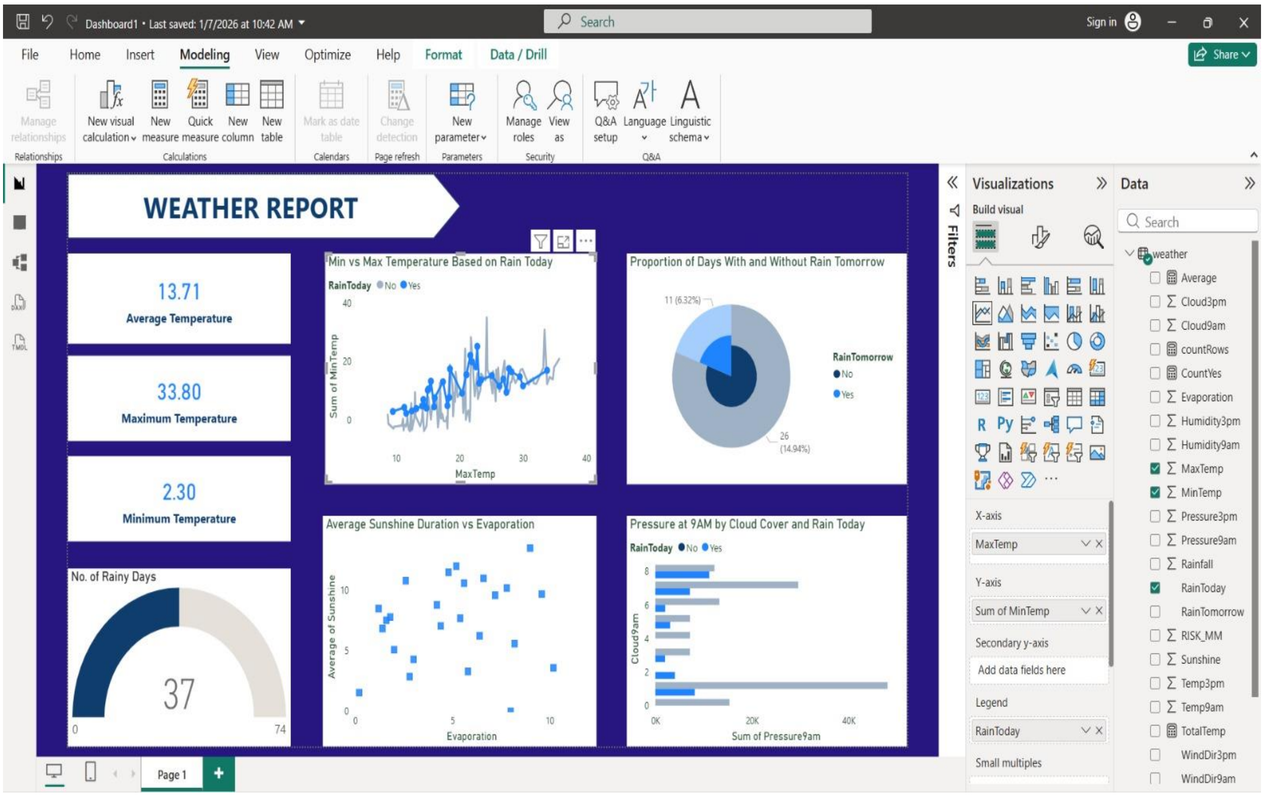

Figure 2

This image shows the analysis of rain in a particular day

CHAPTER 11

CHALLENGES AND LEARNING OUTCOMES

During the development of the Student Performance Dashboard using Power BI, several challenges were encountered while handling the dataset and creating meaningful visualizations. These challenges helped in improving technical skills and gaining practical experience in data analysis tools.

Challenges Faced

- Handling missing values and ensuring the dataset was properly formatted before creating visualizations.
- Choosing the right charts to effectively represent weather data relationships.
- Understanding which variables to use for X-axis, Y-axis and aggregation to create meaningful insights.
- Ensuring correct calculations such as average temperature, maximum temperature, and minimum temperature.
- Arranging visuals in a way that makes the dashboard clear, organized, and easy to interpret.

Learning Outcomes

- Learned how to transform raw weather data into interactive and insightful visual dashboards.
- Gained hands-on experience in creating dashboards using different visuals like KPI cards, scatter plots, gauges, and charts.
- Developed the ability to analyze patterns such as temperature trends, rainfall distribution, and environmental relationships.
- Learned how to design dashboards that are clear, visually appealing, and informative.

CHAPTER 12

CONCLUSION

The Data Analytics internship provided a strong foundation in data analysis techniques using Excel and Power BI. Through this training, important skills such as data cleaning, data visualization, and dashboard creation were learned. The hands-on project helped apply theoretical knowledge to practical scenarios, improving analytical thinking and problem-solving abilities.

Overall, the internship was highly beneficial in understanding how data analytics tools are used in businesses to support data-driven decision making.

CHAPTER 13
FUTURE ENHANCEMENT

In the future, the following improvements can be made:

- Learning advanced Power BI features
- Using Python or R for advanced data analysis
- Implementing machine learning models for predictive analytics
- Working with large real-time datasets.

CHAPTER 14
BIBLIOGRAPHY

1. Microsoft Excel Documentation
2. Microsoft Power BI Official Website
3. Online Tutorials on Data Analytics
4. Training Materials provided during Internship